

MSc 511 --- AOL ASSIGNMENT

- 1) Given the equilibrium on the sheet, what are the price elasticity of supply and the price elasticity of demand at the current price (This is a numerical $+1/-1$ in the price, multiplied by the appropriate ratio)? From a consumer standpoint what kind of good is it?

Using the given equilibrium on the sheet we can see that original price equals= 6,21663 and for that price the demand and supply equal 7149,28.

If we change the price by +0.1 we get a new price that equals 6,31663 and for that price the demand = 7036,1 and supply = 8237,86.

If we change the price by -0.1 we get a new price that equals 6,11663 and for that price the demand = 7266,16 and the supply = 6190,93.

These changes are consistent with topics we covered in lecture. We can see that supply is an increasing function of the price. The change is also consistent with Marshallian demand, that refers to how consumer demand decreases as the price of a good increases. The model follows the law of supply and the law of demand as prices change.

Supply Elasticity = $\frac{\text{change in supply } Q}{\text{change in price}} \times \frac{\text{price}}{Q}$

Demand Elasticity = $\frac{\text{change in demand } Q}{\text{change in price}} \times \frac{\text{price}}{Q}$

The supply elasticity= 8,86346. This value indicates that supply is highly elastic. A 1% increase in price would lead to an 8.86% increase in the quantity supplied.

The Demand Elasticity = -1. That means the demand is unit elastic. A 1% change in price will lead to a 1% change in the quantity demanded, in the opposite direction. This implies that consumers are moderately responsive to price changes, but the total revenue (price $\times$ quantity) remains unchanged as price changes.

Based on our findings I believe that's it's a normal good

- 2) What is the income elasticity in the current equilibrium? What kind of good is this?

Income elasticity = $\frac{\text{Change in demand } Q}{\text{Change in } B} \times \frac{B}{Q}$

I checked the elasticity on the base of changing value of current B (+100 and - 100). I observed the change in Demand quantity in equilibrium. For an 100 increase in B the Demand Quantity increased to 7864,2. For a 900 value of B the Demand quantity fell to the value of 6434,35.

Income elasticity in this case = 1

Based on our findings I believe that's it's a normal good.

- 3) What is the total profit to the industry in this situation?

TOTAL INDUSTRY PROFIT= TOTAL REVENUE – TOTAL COSTS

Total revenue is going to be the equilibrium demand multiplied by the price. So, if we multiple 7149,227 (our demand) by 6,216635 (our price) we get 44444,44 in total revenue.

Total cost is the cost of producing W1 quantity for Firm 1 and cost of producing W2 quantity for Firm 2. W1 and W2 quantities is the amount each company produce out of the total equilibrium supply.

Total cost= Cost of producing 6888,501 units for Firm 1 + Cost of producing 260,7758 units for Firm 2 = $38540,97 + 1377,976 = 39918,94$.

Total Industry Profit= $44444,44 - 39918,94 = 4525,503$

Total Industry Profit equals \$ 4525,503.

4) What is the Profit of each firm?

To count each Firms individual profit, we need to calculate their individual profits and use individual cost from previous point. For Firm 1 its individual costs equal 38540,97 and for Firm 2 individual costs equal 1377,976.

To find individual revenue for each firm we multiple W1 and W2 quantities - the amount each company produce out of the total equilibrium supply by the equilibrium price.

For Firm 1 their revenue = $6888,501 * 6,216635 = 42823,3$

For Firm 2 their revenue = $260,7758 * 6,216635 = 1621,148$

Total profit of Firm 1 and Firm 2 depends on individual revenue and their individual cost. We found the cost in previous point by putting their share in supply quantity W1, W2 into W column for both companies and thanks to the formula we got the cost of production for each firm. We counted their individual revenue Now that we have both the revenue and cost for both firms, we can calculate profit for Firm 1 and Firm 2.

For Firm 1, Individual Profit= $42823,3 - 38540,97 = 4282,331$

For Firm 2, Individual Profit = $1621,148 - 1377,976 = 243,1722$

5) SCENARIO A...make a copy of your supply-demand sheet and look at the following scenario:

The Demand function shifts because 30 more people come into the market (same utility).

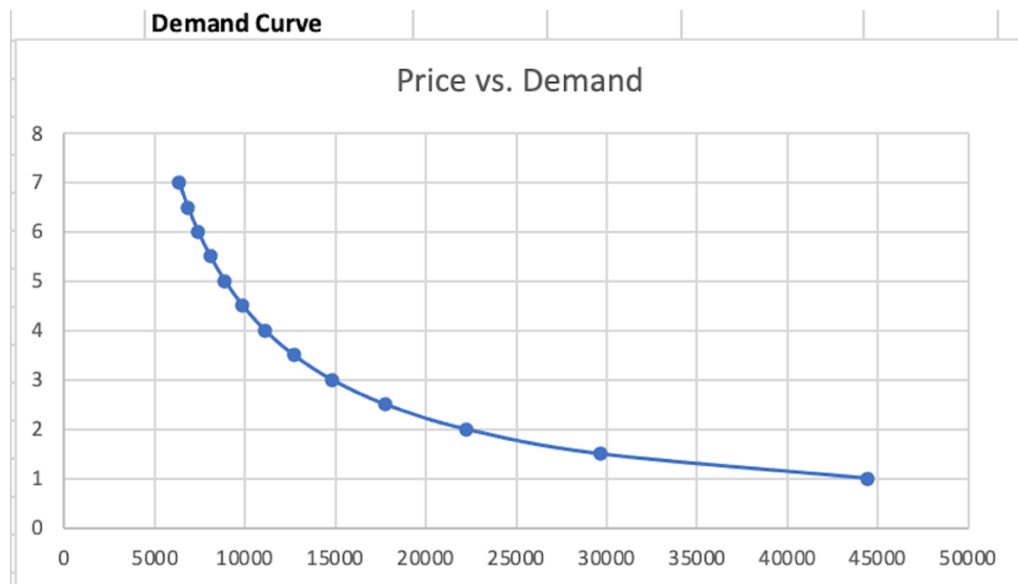
How much does price and quantity change? In terms of total demand does this make sense with respect to the elasticities?

We go the Demand table, and we change the number of individuals from 100 to 130. After that we can clearly see a change the Demand curve. It goes from Curve 1 to Curve 2 after the change in the number of individuals.

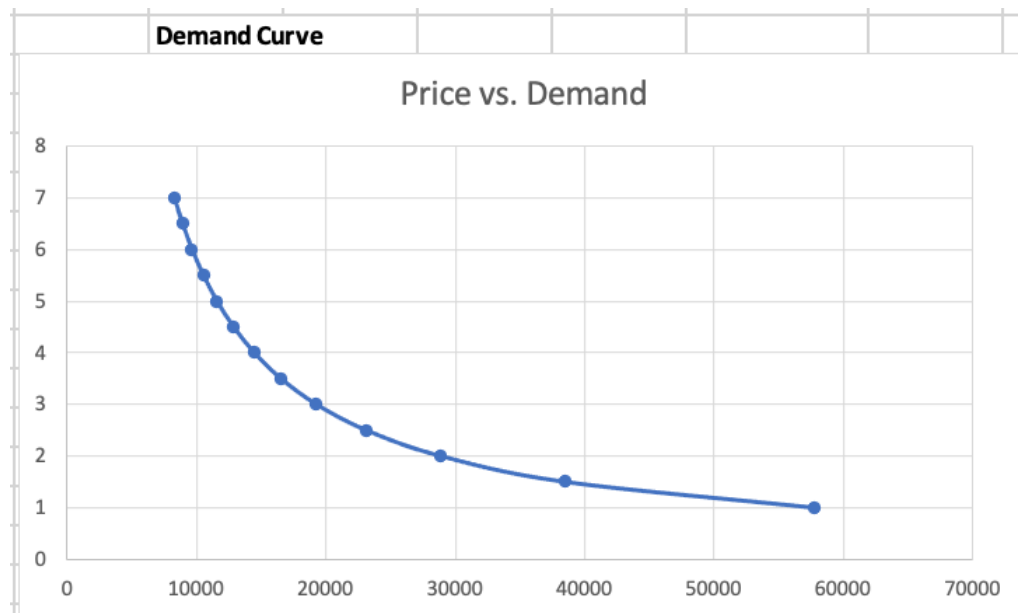
This directly impacts the total demand at each price point since the larger population (130 individuals) generates a higher total demand compared to the smaller one (100 individuals). In the first Demand Curve total revenue is lower across all price points because less individuals contribute to the demand. However, on the second curve total revenue is higher- due to the increase in the population size. Both curves exhibit the same general shape, reflecting—higher prices lead to lower demand. The scale of the x-axis (demand) differs, with the second curve extending further due to the higher number of individuals. With the change in number of individuals our demand is higher so supply and demand don't match anymore. With solver we match them and find out our new equilibrium price. Our new

equilibrium quantity so amount at which supply, and demand intersect is: 9050,588 with that our new equilibrium price is 6,38387.

Curve 1:



Curve 2:



It does not match our elasticities entirely. With Demand Elasticity = -1 and Price Elasticity=1 the change happening in this example is not exactly consistent with our previous findings. Elasticity of -1 means that a 1% change in price results in an exactly 1% change in demand quantity (in the opposite direction). If price increases, quantity decreases proportionally, so revenue stays the

same. If price decreases, quantity increases proportionally, and revenue still stays constant. Total revenue remains constant regardless of changes in price along a single demand curve. In this specific scenario because the population increased, so did the demand and then the price for equilibrium, we also see a change in total revenue. So, while some findings are consistent this part does not match our found elasticities.

- 6) SCENARIO B...make a copy of the Supply-Demand sheet...now, a third firm comes in with a production function given by $\alpha=.45$ $\beta=.4$. What happens to the equilibrium price, the total supplied, and the profit of each firm?

We create the table for a new, third company. It is structured identically as tables for Firm 1 and Firm 2. We assign $\alpha=0,45$ and $\beta=0,4$ to the Firm 3. Rest of the values such as w_1 , w_2 stay the same for clarity. We get all the values for Firm 3 using the same formula as in tables for other two firms.

What we need to find a new equilibrium is: W and K value for third company. We get $K_3=2,088128$ thanks to our α_3 and β_3 values. With K_3 we can calculate $W_3=189,763$. When we add $W_1+W_2+W_3$ we get our new supply quantity, which is 7168,72. Using solver we match the demand to the supply value and get our new equilibrium price, 6,199774.

With a third firm coming in, we saw an increase in total supplied quantity and a decrease in equilibrium price.

We also see a drop in relative shares (W_1, W_2, W_3). Factors of production change. Logically with new company entering a market, we saw a new higher supply and a new lower price. The profits of Firm 1 and Firm 2 will decrease, but a small one. Firm 1 saw a decrease of around 2,68% while Firm 2 experienced a 1,79% loss in profits.

Firm 3 Individual Profit = $1176,493 - 1000,02 = 176,4728$

- 7) If firms keep entering the market what would be the long run equilibrium profit, and can you tell which of these three types of firms would dominate?

The price of the good will continue to decrease as more firms enter the market, BUT not all companies share the same cost production structure. As additional firms join, the price will decline until it reaches the production cost of the firm with the lowest production cost. However, since production costs vary among companies, the firm with the lowest production cost is likely to dominate the market. Among the three firms analyzed, Firm 1 stands out as the type that will dominate the market. Why? Because in that scenario the market will be dominated by the company that as the price of the good goes lower and demand increases will keep their production cost of the demand quantity as low as possible.

While Firms 2 and 3 have lower production costs than Firm 1 at low output levels, Firm 1 achieves significantly greater cost efficiency at higher production scales. For instance, at a production level of 10,000 units, firm 1's cost per unit is approximately half that of Firms 2 and 3.

Industry will go as low as at least one firm is making profit or its profit = zero. There can't be a negative profit for the market. But clearly one of the companies will have lower cost of production than others. This company will have the market as it has the ability to go lower than the other firms.